



AAMA 503-24

Voluntary Specification for Field Testing of Newly Installed Storefronts, Curtain Walls and Sloped Glazing Systems

an FGIA publication

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HOW TO USE THE SHORT FORM SPECIFICATION

To standardize and simplify the writing of field testing specifications for storefronts, curtain walls and sloped glazing systems, FGIA has prepared the following "Short Form Specification" for use by architects/specifiers. It may be used by merely inserting the following paragraph(s) into the project specifications. Please note that "Specifier Notes" in *italics* are not to be inserted into the project specifications.

SHORT FORM FIELD TESTING SPECIFICATION

1. The newly installed (*) shall be field tested, as contractually agreed upon by the interested parties, by an AAMA accredited independent laboratory, as engaged by (), in accordance with AAMA 503, "Voluntary Specification for Field Testing of Newly Installed Storefronts, Curtain Walls and Sloped Glazing Systems."

* *Insert storefront, curtain wall and/or sloped glazing system.*

SPECIFIER NOTE 1: *See Section 5.0 for a description of the test method. AAMA 503 specifies air leakage resistance and water penetration resistance field testing of curtain walls, store fronts, and sloped glazing systems during construction and prior to issuance of the building occupancy permit.*

2. The area(s) to be tested is (are) as follows:

**

** *Insert an exact description of the area(s) to be tested. Failure to do so can dramatically affect the cost of the field testing.*

SPECIFIER NOTE 2: *The recommended procedure for identification of the test specimen is to reference an architectural elevation/plan drawing that clearly shows the intent of the area to be field tested. The specimen(s) to be tested on a project should be determined after careful consideration of the following factors:*

- *The quantity, location and size of the specimen(s) selected for testing can markedly affect the cost of testing. (See Section 6.4.)*
- *Selecting a large number of specimens (or large area(s) to be tested) on a small project (less than 93 m² (1000 ft²) of glazing) is normally not cost effective and as few as one specimen (or surface area as small as 9.3 m² (100 ft²) of glazing) may provide the information needed.*
- *On all projects, regardless of size, the specifier should determine how many specimens and/or their desired surface area, are required in order to establish a reasonable measure of quality for the entire project. ASTM E122 provides guidance on how to establish the number of test specimens in order to estimate a measure of quality of a production lot with prescribed precision.*
- *The specifier should carefully balance the cost of the specimen or specimen preparation, as well as the cost of testing and the restoration of the project surrounding components and finishes to their original condition, before determining the number, location and size (surface area) of specimens to be tested. On larger projects, a formal cost-benefit analysis is appropriate. On smaller projects, the specifier should avoid a situation where the cost of testing and building restoration approaches the cost of the glazing system(s). If any of the tested specimens fail to comply with the project requirements, considerations should be given to the selection and testing of additional areas.*
- *In the event of observed water penetration, if the source of the penetration cannot be determined, a forensic evaluation using the procedures outlined in AAMA 511 while maintaining the test pressures and methods defined in AAMA 503 will be required.*

SPECIFIER NOTE 3: *See Section 7.0 for a description of the test method and default performance specifications. AAMA 503 specifies water penetration resistance field testing for the above area immediately after installation begins. Any of the following paragraphs may be added to modify the standard specification:*

3 Air leakage resistance tests shall be conducted at a uniform static test pressure of Pa (psf). The maximum allowable rate of air leakage shall not exceed L/s•m² (cfm/ft²).

SPECIFIER NOTE 4: *Field air leakage resistance tests are only recommended for discrete "punched opening" curtain walls, storefronts and sloped glazing elevations. Field air leakage testing is not recommended for a portion of a continuous system due to the problems associated with compartmentalization and identification of extraneous air leakage. Refer to Sections 7.1 & 7.2 for additional precautionary information and recommended performance criteria.*

4. Water penetration tests shall be conducted at a static test pressure of Pa (psf).

SPECIFIER NOTE 5: *Unless stated otherwise and clearly stipulated in the project specifications, in no case shall the specified static water test pressure differential exceed 2/3 of the tested or rated laboratory performance (see Section 7.3 for default performance requirements). If any test specimen fails to meet the requirements, refer to AAMA 503 Section 7.3.7.*

0.0 INTRODUCTION

0.1 The purpose of this specification is to provide a method to evaluate the installed performance of newly installed storefronts, curtain walls and sloped glazing systems and their installation during construction for air leakage resistance and/or water penetration resistance under controllable, reproducible and appropriate conditions. This is for the purpose of providing reasonable assurance that as installed, the fenestration products will perform to the specified or rated air leakage and/or water penetration performance levels with allowances given for field performance.

0.2 This specification establishes the requirements for field test specimens, apparatus, sampling, test procedures and test reports to be used in verifying the air infiltration resistance performance and/or water penetration resistance performance of newly installed fenestration products.

1.0 SCOPE

1.1 These specifications establish the requirements for test specimens, apparatus, sampling, test procedures and test reports to be used in evaluating the performance of newly installed storefronts, curtain walls and sloped glazing systems and their installation during construction, prior to issuance of the building occupancy permit, but no later than six months after issuance of the occupancy permit.

1.1.1 Specifically excluded from this specification are products within the scope of AAMA/WDMA/CSA 101/I.S.2/A440 (NAFS); these products shall be tested per AAMA 502 as required. Also excluded from this specification are commercial entrance systems. Verifying the structural loading resistance performance of fenestration products is beyond the scope of this specification.

1.2 Unless otherwise agreed upon, if field testing is required more than six months after the building occupancy permit has been issued, AAMA 511 shall be used. Additionally, in the event that water penetration testing using the test methods outlined in this specification produces water penetration that cannot be solely attributed to the glazing system, forensic testing shall be initiated by using AAMA 511 and ASTM E2128. The determination of the source(s) of water penetration from a wall or roof component other than the glazing system requires forensic investigation and testing outside the scope of this specification.

1.3 In this specification, “shall” is used to express a requirement, i.e., a provision that the user is obliged to satisfy in order to comply with the specification; “shall be permitted to be” is used to express an option or that which is permissible within the limits of the specification; “should” is used to express a recommendation or that which is advised but not required; and “may” is used to express possibility or capability. Notes accompanying sections do not include requirements or alternative requirements; the purpose of a note accompanying a section is to separate explanatory or informative material from the text.

Notes to tables and figures are considered part of the table or figure and shall be permitted to be written as requirements. Section 0.0 and any Notes not attached to figures and tables are non-mandatory.

1.4 The primary units of measure in this document are metric. The values stated in SI units are to be regarded as the standard. The values given in parentheses are for reference only.

1.5 This document was developed in an open and consensus process and is maintained by representative members of FGIA as advisory information.

2.0 SIGNIFICANCE AND USE

2.1 This document provides a method which shall be used to evaluate the performance of newly installed storefronts, curtain walls and sloped glazing systems and their installation during and just after initial construction for air leakage rate and/or water penetration resistance under controllable, reproducible and appropriate conditions.

2.2 The specifier/architect shall use this document and the short form provided to communicate to the fenestration product manufacturer the installed performance for air infiltration and/or water penetration. Following product installation, this document provides a structured, controllable and reproducible method for testing.

2.3 Improper use of this document may lead to a misunderstanding of a fenestration product’s performance and/or the misapplication of fenestration products on the project.

2.4 The use of this specification for the testing of installed fenestration products presumes that all personnel comply with the requirements of AAMA LAP-3, *Laboratory Accreditation Program Operations Manual- Laboratories and Test Agencies Performing On-site Testing of Fenestration Products*. These requirements include, but are not limited to, a working knowledge of the following:

2.4.1 Building construction and fenestration installation techniques.

2.4.2 The various fenestration types and designs, their operation and application.

2.4.3 Understanding how the placement and attachment of air and water test chambers can affect test results.

2.4.4 The understanding of how to isolate leakage through system components such as fenestration trim/panning, perimeter and other extraneous leakage.

2.4.5 How interior and exterior ambient conditions (temperature, wind, barometric pressure and moisture) affect measurement accuracy and how they affect the specimen being tested.

2.4.6 The principles of pressure measurement, fluid mechanics and instrumentation practices.

2.4.7 Assurance that on-site verifications are conducted in accordance with Section 8.0.

2.5 All testing shall be conducted by an AAMA accredited laboratory or field test agency. See AAMA LAP-3 for more details.

3.0 REFERENCED DOCUMENTS

3.1 References to the standards listed shall be to the edition indicated. Any undated reference to a code or standard appearing in this section shall be interpreted as referring to the latest edition of that code or standard.

3.2 AAMA, Fenestration and Glazing Industry Alliance (FGIA) Standards

AAMA/WDMA/CSA 101/I.S.2/A440-22, North American Fenestration Standard/Specification for windows, doors, and skylights

AAMA LAP-3-17, Laboratory Accreditation Program Operations Manual- Laboratories and Test Agencies Performing On-site Testing of Fenestration Products

AAMA 501.1-17, Standard Test Method for Water Penetration of Windows, Curtain Walls and Doors Using Dynamic Pressure

AAMA 502-21, Voluntary Specification for Field Testing of Newly Installed Fenestration Products

AAMA 511-22, Voluntary Guideline for Forensic Water Penetration Testing of Fenestration Products

FG-22, FGIA Glossary

3.3 ASTM International (ASTM)

ASTM E122-17(2022), Standard Practice for Calculating Sample Size to Estimate, With Specified Precision, the Average for a Characteristic of a Lot or Process

ASTM E283/E283M-19, Standard Test Method for Determining Rate of Air Leakage Through Exterior Windows, Skylights, Curtain Walls, and Doors Under Specified Pressure Differences Across the Specimen

ASTM E783-02(2018), Standard Test Method for Field Measurement of Air Leakage through Installed Exterior Windows and Doors

ASTM E1105-15, Standard Test Method for Field Determination of Water Penetration of Installed Exterior Windows, Skylights, Doors, and Curtain Walls, by Uniform or Cyclic Static Air Pressure Difference

ASTM E2128-20, Standard Guide for Evaluating Water Leakage of Building Walls

4.0 DEFINITIONS

Please refer to the most current FGIA Glossary for all definitions.

4.1 In the industry “test specimen” and “test area” are used interchangeably. For the purposes of this specification, the term “test specimen” is used.

5.0 TEST METHODS

5.1 Field testing procedure and test apparatus shall meet the requirements of the following referenced ASTM test method.

5.1.1 Resistance to air infiltration using static air pressure difference: ASTM E783.

5.1.2 Resistance to water penetration using static air pressure difference: ASTM E1105.

NOTE 1: AAMA 501.1 dynamic testing is sometimes used for field (onsite) testing, especially for large projects and when construction of a chamber is not feasible. AAMA 501.1 is currently written to apply to both lab and field testing. However, AAMA 501.1 is limited to testing protocols and provides limited guidance for specifying testing quantities, system capability, pressures, etc. The specifier is encouraged to review those portions of AAMA 503 addressing those issues.

5.1.3 The AAMA accredited test agency shall ensure these test methods, AAMA 503 and applicable project specifications are available for review at the test site upon request, subject to copyright restrictions.

5.2 Test Chamber Arrangement

5.2.1 Joints between the specimen and the adjacent wall, curb or roof shall be permitted to be included in the test specimen unless otherwise noted in the project specifications. The test chamber shall be applied to the interior or exterior of the wall, curb or roof construction in such a manner as to create a pressure differential across the specimen assembly, including perimeter frame intersections and perimeter seals, subframes, receptors and flashing.

It is not practical to install a chamber on a segment of a continuous horizontal or vertical member; therefore, air leakage testing of these systems is not recommended.

5.2.1.1 This test method ensures that the entire test specimen (including all glazing systems and associated frame members, joints, subframes, receptors, flashing, etc.) and the adjacent substrate, including the perimeter seals, are all tested under the prescribed pressure differential. The test shall be performed soon after the specimen is installed, sealants are cured and prior to the installation of gypsum wall board, insulation or other finish materials. If interior wall materials have been installed, they shall be removed at the test specimen to allow for visual access to these areas to check for water leakage. The test chamber shall be installed to the specimen in such a manner as to apply a pressure differential to all joinery conditions (including perimeter seals) and so as to minimize extraneous air leakage.

NOTE 2: Wind and other environmental conditions (barometric pressure, rain and temperature changes) can have an adverse impact on field testing performance; therefore, the specimen should be isolated from adverse ambient conditions with a barrier, and conditioned, if necessary, to protect it from any environmental condition which may affect the test results. If testing is performed under conditions that would affect the test results, such as extreme temperatures, high winds or other adverse environmental conditions, these conditions are noted in the report. Performance of any test without appropriate consideration and adjustment for ambient conditions may render the test results invalid.

5.2.2 In the case of a stud wall/roof system, the test chamber perimeter line, if installed on the building interior, would normally be attached to studs on both sides of the specimen and at the head and sill framing.

5.2.3 On a precast wall/roof system, a chamber installed on the interior can be sealed directly to the back of the precast panel.

6.0 SAMPLING, SITE PREPARATION AND RESPONSIBLE PARTIES

6.1 As soon as practical after installation has begun, and after a representative test specimen has been completely installed, adjusted, cleaned and perimeter sealed (including adequate time for the sealant to cure), the specimen(s) shall be tested as specified in Section 7.0 for air leakage rate and/or water penetration resistance.

6.1.1 The test proponent shall notify the fenestration product installer, manufacturer and the responsible party defined in Section 6.2, of the test schedule. The initial advance notice shall be a minimum of two weeks in advance of the testing in order for the installation general contractor to notify all fenestration product trades (i.e., erector, glazier, perimeter caulk contractor, etc.) of the scheduled test date.

NOTE 3: *Performing the field test as soon as practical may be beneficial in determining if manufacturing, installation and/or perimeter sealing problems are present before a substantial portion of the project is completed. If the initial testing is performed early in the project installation process, necessary corrections can be made with less financial impact or other deleterious effects on the project. On large projects, tighter construction monitoring may be performed by testing of installed products at approximate intervals of 5%, 50% and 90% completion of the installation.*

6.2 The project specifications and/or the contract documents shall clearly identify the responsible party who is liable for the costs associated with the initial field testing, retesting and, if necessary, forensic testing.

6.3 Testing shall be performed by an AAMA accredited laboratory or field testing agency.

NOTE 4: *AAMA accredited laboratories or field testing agencies are required to comply with AAMA LAP-3. The requirement of AAMA accreditation assures the specifier that the agency has the staff, training, experience and calibrated equipment to properly perform field testing.*

6.4 The test specimen(s) shall conform to the following:

6.4.1 The test specimen's size and location shall be selected by the architect and shall be clearly identified within the construction documents. If the specimen location has not been pre-determined, the location shall be selected by the architect or owner's representative unless delegated to the AAMA accredited testing agency or building envelope consultant.

6.4.2 If no specimen size and/or location is identified in the construction documents, select an appropriate-size specimen that will provide representative performance data. A single 9.3 m² (100 ft²) test specimen is typically the recommended specimen when contained within a larger wall area. This recommended test specimen does not apply to punched openings less than 100 ft². If more than the default size and/or quantity is desired, the number of specimens, area and intervals of testing shall be stipulated in Paragraph 1 of the Short Form Field Testing Specification. The specimen shall include perimeter seals, typical splices, frame intersections and, if applicable, at least two entire vision lites and two entire spandrel lites containing an intermediate vertical and an intermediate horizontal. (See Figures 1, 2 and 3 for typical test assemblies.)

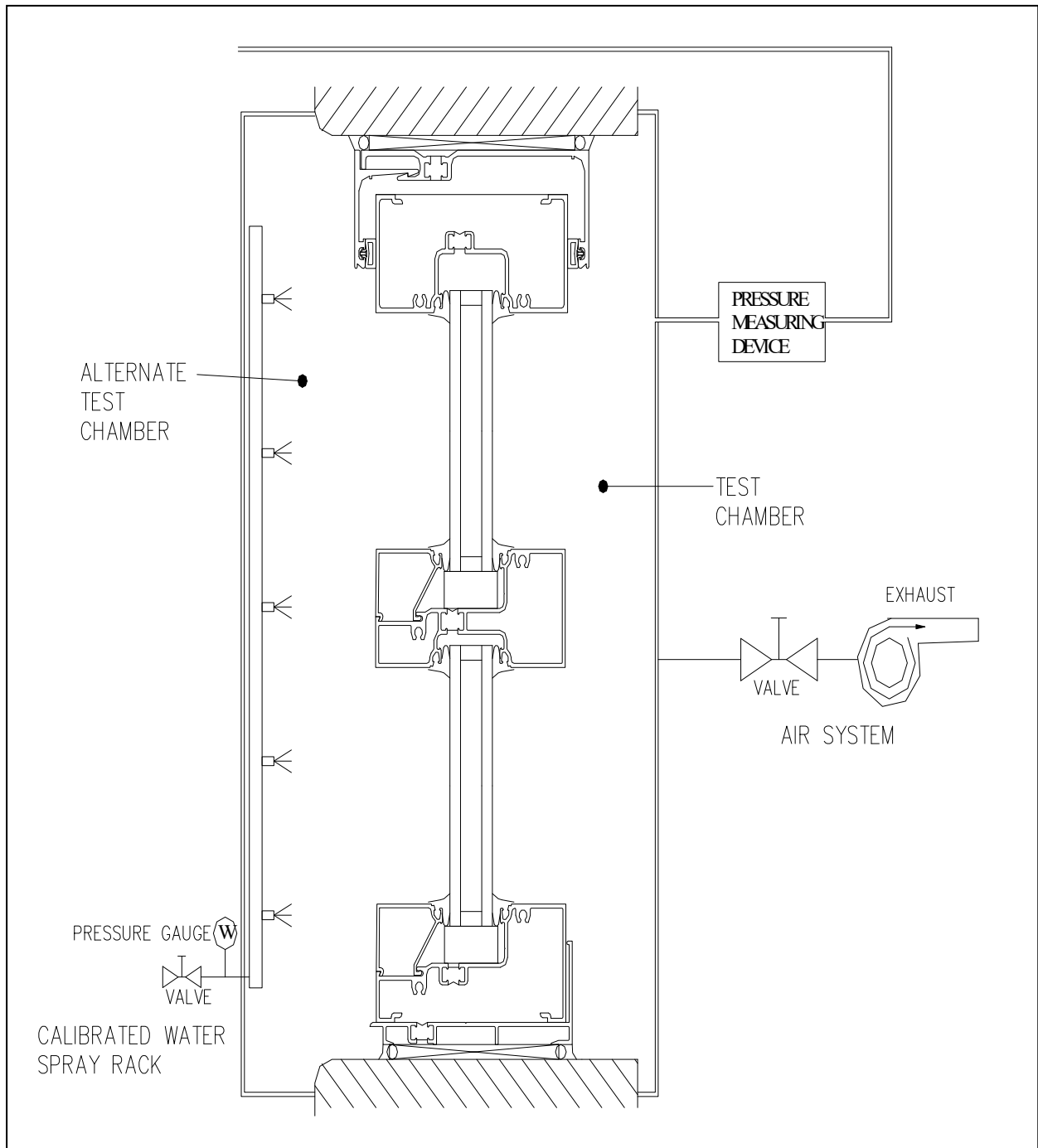


Figure 1: Test Chamber Example- Storefront

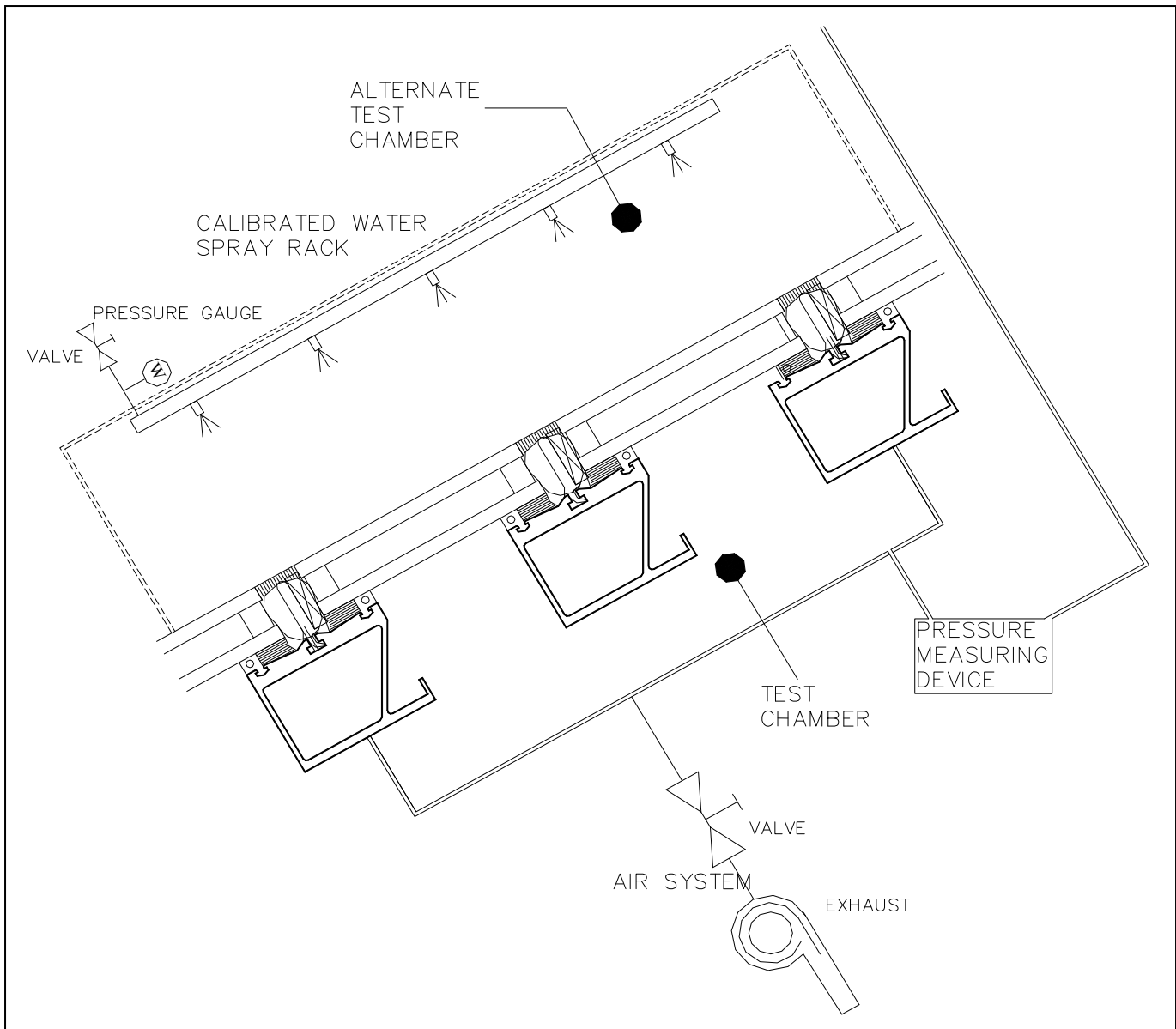


Figure 2: Test Chamber Example- Sloped Glazing

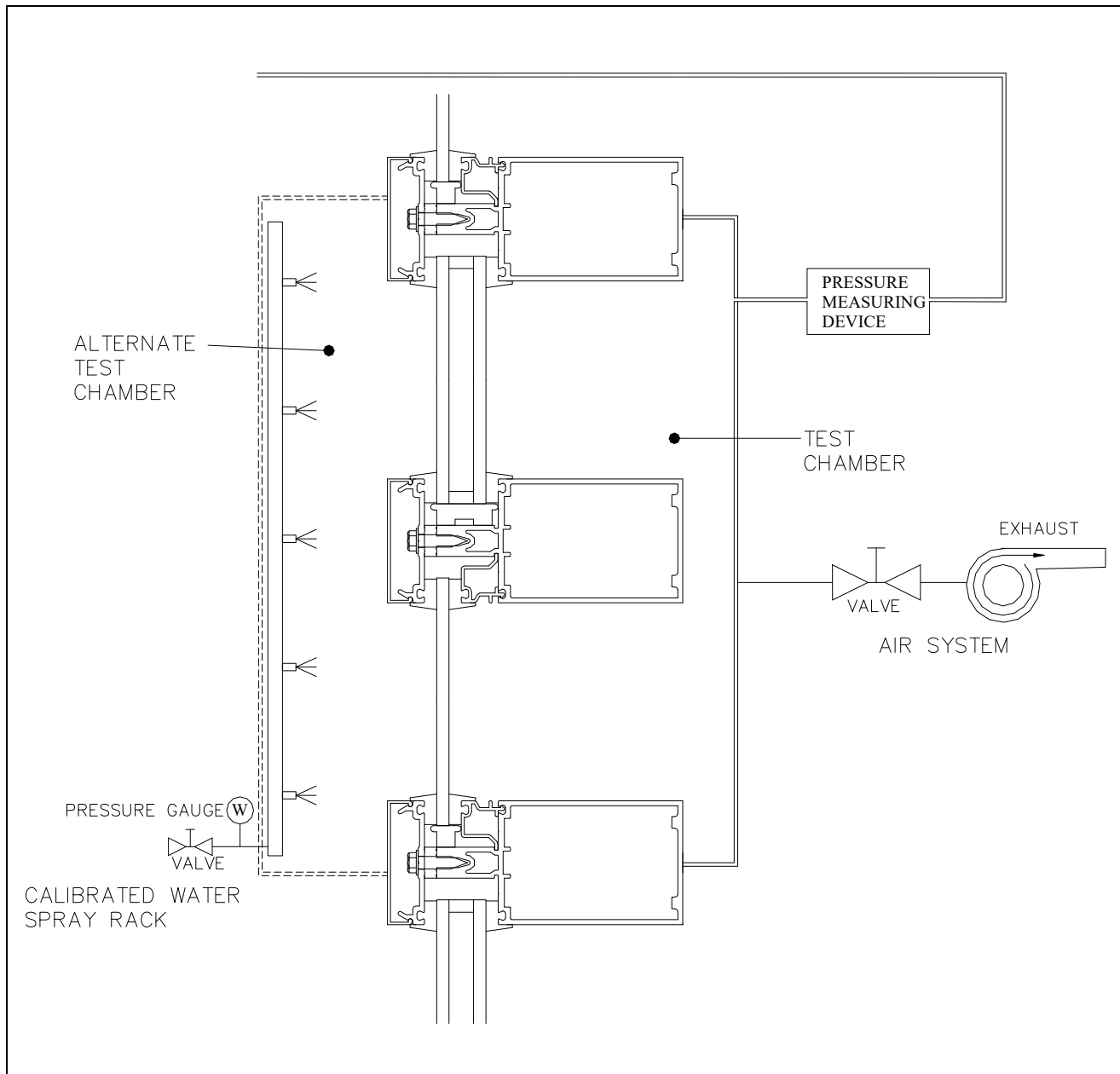


Figure 3: Test Chamber Example- Curtain Wall

6.4.2.1 If the specimen contains an integral ventilating system/device (refer to AAMA/WDMA/CSA 101/I.S.2/A440-22, Clause 8.2.1), it shall be permitted to be tested twice for water penetration resistance and twice for air leakage resistance, once with the venting portion of the ventilating system/devices in the closed position and again if necessary with the venting portion closed and taped or sealed. Air and/or water performance values for both test modes shall be included in the test report.

6.4.3 The test specimen shall be representative of typical installations and construction for the project. The specimen(s) shall have no outstanding punch list items or visible damage or irregularities, nor be singled out because of obvious performance problems.

6.4.3.1 Unless otherwise specified, exterior cladding, rain screen and associated components, including the exterior perimeter sealants, shall be installed within and/or adjacent to the fenestration product to represent the final detailed conditions of the exterior envelope.

6.4.3.1.1 If it is desired to test the air and vapor barrier and seals adjacent to the fenestration product prior to the exterior wall cladding being installed, the fenestration specimen perimeters shall be protected during that testing and the fenestration product shall not be evaluated. The same protection could be used to simulate the cladding and perimeter seals, should subsequent testing of the fenestration product be desired. See Figure 4.



FIGURE 4: Exterior Wall Cladding and Associated Components Applied

6.4.4 After the specimen(s) locations have been selected, the owner's representative shall direct the responsible contractor and manufacturer representative to remove interior finishes (if necessary), clean the specimen(s), to assure that the specimen is installed in accordance with the fenestration manufacturer's installation instructions and remove adjoining sheetrock, trim, insulation or other materials, which could adversely affect the chamber attachment and visual observation. Care shall be taken not to disturb the interior air seal, if present. Interior or exterior components, sealants, etc., that are required for product performance shall not be removed as some product installations require an interior air seal to perform as designed. A means to view, observe, or otherwise identify test-related water penetration shall be accommodated by the contractor if requested by any responsible party, provided that the appropriate parties are notified that repairs must be made.

It is recommended that visual access be provided before the testing to verify that the test specimen does not contain residual water prior to testing. If visual access is provided at the conclusion of testing, any observed water meeting the definition of water penetration shall be considered to have entered during the test.

The responsible contractor shall coordinate the supply of sufficient power and water per the requirements of the AAMA accredited test agency.

The designated specimen(s) shall be inspected by the testing agency. The pretest inspection shall include the checking and recording of the following in the test report:

- The plumb, level and square of the specimen.
- The operation of the specimen (if applicable).
- The pre-test condition of all surfaces to confirm that they were in a condition where a judgment can be made as to whether water penetration is a result of this testing.

The fenestration manufacturer and the installer shall also be afforded the opportunity to perform an inspection prior to testing. The manufacturer and the installer shall be allowed to correct any deficiencies prior to testing. If the deficiencies cannot be corrected another sample shall be selected.

If the selected test locations are not easily and/or safely accessible from both the inside and outside, significant additional cost may arise for scaffolding, swing stage, electrical and water sources. If identical products are installed at more accessible locations, these locations shall be selected for testing. Whenever possible, the architect/specifier shall define the field-testing requirements in the project specifications, including the number of tests and their location. This will provide some assurance that consideration has been given to the complexity of the field testing prior to bidding and further communicate the requirements to those who will be responsible for, or involved in, the field testing.

6.4.5 Upon completion of the testing, the responsible contractor shall replace or repair items removed in Section 6.4.4.

7.0 TEST PROCEDURES

7.1 Air leakage and/or water penetration resistance tests shall be performed at pressures specified in Sections 7.2.3 through 7.3.1 unless stipulated otherwise in the Short Form Field Testing Specification. Where both tests are to be conducted in sequence, the test for air leakage resistance shall be conducted before the test for water penetration resistance.

7.1.1 The testing agency shall issue a field test plan to the entity or party contracting the test agency prior to testing that summarizes a specification review, test methods/standards, failure criteria, identifies products to be tested, number of products to be tested, locations, etc.

7.1.2 If there is reason to believe that residual water from rain or other sources may be located in the specimen, a negative (outward) pressure test followed by a positive (inward) pressure test shall be conducted at the same pressure differential used for the performance test to purge the specimen of any residual water. The specimen gaskets or weatherstrips shall be examined and shall be dry before proceeding with the air leakage resistance test.

7.1.3 Other than as specified in Section 7.2.2, test chambers and their associated components shall not be supported at any point by the test specimen. (See Figures 1, 2, and 3.)

7.1.4 Consideration shall be given prior to the application of water to assure that the desired water test pressure is achieved.

7.1.5 Consideration shall be given to placement of pressure taps in relationship to the blower inlet of the chamber, as well as mullions, meeting rails/stiles and other joints. Pressure taps should generally be placed near the center of the product.

NOTE 5: *The pressure measuring device shall take into account the difference that may exist between the static pressure in the interior of the building and the static pressure on the exterior. It may be possible to open a window or door in a room to balance the pressure. If this is not possible, a measurement of the outdoor wind speed and/or indoor building pressure shall be made and the test pressure adjusted accordingly.*

7.2 Air Leakage Rate

7.2.1 An air leakage resistance test shall be conducted at a minimum uniform static test pressure of 75 Pa (1.6 psf) or as specified for the project, but not to exceed 300 Pa (6.2 psf). Air leakage resistance (when required) shall be determined and adjusted to standard conditions per ASTM E 783.

7.2.1.1 If changes in outdoor wind speed and/or indoor air pressure are sufficient to cause static air pressure difference readings to fluctuate more than $\pm 10\%$ from the specified test pressure, one of the following methods shall be used:

- Method A: The specimen and/or test chamber shall be shielded or temporarily enclosed to minimize these fluctuations.
- Method B: Record 10 readings of air flow, evenly distributed over a 30- minute period for each of the total and tare leakage measurements. Report both the average of the ten readings and standard deviation in the test report.
- Method C: Equipment used to maintain the pressure differential during testing shall continuously monitor test pressure and self-adjust to maintain the appropriate test conditions.

NOTE 6: *Flexible materials are sometimes used in the construction of the test chamber. Care should be taken to stabilize the differential chamber pressure prior to recording all measurements. Large chambers in particular require sufficient time for volume adjustment to occur prior to recording the measurement.*

7.2.2 Extraneous Air Test

Measurement of extraneous air leakage (tare) readings require special consideration on many on-site installations. A review of installation details to identify sources of tare leakage shall be conducted prior to test commencement. Precautions shall be taken to minimize the extraneous air leakage. Refer to Section 11 of ASTM E783). If the extraneous air leakage exceeds the limits specified in ASTM E783, then the chamber may be attached to the perimeter frame of the test specimen for the purposes of obtaining an accurate tare reading.

The intent of the air leakage test is to measure only exterior air leakage through the joints, fixed glazing, hardware, etc., and within the area enclosed by the specimen frame. The air is assumed to be coming from the exterior of the building directly through the specimen. Other sources of air infiltration (i.e., through fastener holes, interior trim and interior mullion covers, exterior panning systems, receptors, flashing and through leakage in the testing equipment or air chamber attachment to the specimen) are referred to as extraneous air leakage and must be accounted for. It is not practical to account for all sources of air leakage when testing a portion of a continuous curtain wall or sloped glazing. For this reason, it is not recommended that field air leakage tests be performed on this type of system. If the specifier has any doubt regarding the validity of a field air leakage test, he should consult with the independent test laboratory or the glazing contractor. (Only required if initial air leakage value exceeds the allowable value.)

7.2.3 The maximum allowable rates of air leakage for field testing shall not exceed 1.5 times the project specification rate, or $0.5 \text{ L/s}\cdot\text{m}^2$ (0.09 cfm/ft^2), whichever is greater. The specifier may require project specific air leakage; however, this must be stipulated in Paragraph 2 of the "Short Form Field Testing Specification" or in the project specifications.

NOTE 7: *The purpose of in-field air leakage testing is to provide reasonable assurance that the fenestration product will perform to its specified or rated level with allowances given for field reductions. The allowable air leakage used for tests conducted in the field are not the same as the laboratory allowance. This is to allow for field conditions and test methods that vary from the laboratory test conditions and test methods. These conditions are primarily related to the ambient environmental conditions and the installation. The rated product performance is based on laboratory testing performed under controlled laboratory conditions. The temperature, wind and barometric pressure conditions during a field test will typically vary from the standard laboratory conditions.*

The field installation conditions also influence the product performance. Products tested in the laboratory are perfectly plumb, level and square in a precision opening. Field test specimens, although installed within acceptable industry tolerances, are rarely perfectly plumb, level and square. Shipping, handling, acts of subsequent trades, aging and other environmental conditions all may have an adverse effect upon the performance of the installed specimen. A 50% increase in the allowable air leakage for field testing is specified as a reasonable adjustment for the differences between a laboratory test environment and a field test environment.

7.3 Water Penetration Resistance

7.3.1 Water penetration resistance performance shall be determined per ASTM E 1105 using Procedure A, "Uniform Static Air Pressure Difference." The field water penetration resistance tests shall be conducted at a static test pressure of two-thirds of the specified project water penetration test pressure, but not less than 200 Pa (4.18 psf). The specifier may increase the field water test pressure to the value specified for the project; however, this should be stipulated in Paragraph 2 of the "Short Form Field Testing Specification" or the project specifications.

NOTE 8: In the event that the project does not have a specified water penetration resistance test pressure, the value should be equal to the manufacturer's published laboratory test pressure (refer to 7.3.6.b) or 20% of the positive ASD design wind load, whichever is greater, times 0.667.

NOTE 9: The default pressures used for water penetration resistance tests conducted in the field are not the same as the laboratory test pressure to allow for field conditions and test methods that vary from the laboratory test conditions and test methods. These conditions are primarily related to the ambient environmental conditions and the installation. The product performance is based on laboratory testing performed under controlled laboratory conditions. The temperature, wind and barometric pressure conditions during a field test will typically vary from the standard laboratory conditions.

The field installation conditions also influence the product performance. Products tested in the laboratory are typically installed near-perfect for plumb, level and square within a precision opening. Field test specimens, although installed within acceptable industry tolerances, are rarely perfectly plumb, level and square. Shipping, handling, acts of subsequent trades, aging and other environmental conditions all may have an adverse effect upon the performance of the installed specimen. A 1/3 reduction of the test pressure for field testing is specified as a reasonable adjustment for the differences between a laboratory test environment and a field test environment.

7.3.2 Prior to each test, a visual confirmation shall be made that all nozzles are flowing freely and that the straightness and angularity of the nozzles is normal to the plane of the test specimen. Any non-functioning nozzles shall be repaired or replaced.

7.3.3 During the test period, a laboratory or test agency representative shall visually confirm that all test surfaces are uniformly and sufficiently wetted. For test samples larger than the calibrated rack, multiple spray racks may be used provided that the calibration for each rack has been performed as described previously and a sufficient water source is used to supply the pressure established during calibration as recorded in Section 8.3.

7.3.4 In the event the water source is not sufficient to supply the entire test specimen at the calibrated pressure it is permissible to test smaller sections of the test specimen with smaller water spray racks. The test report shall note this exception and any influence on the test results. The specimen shall be allowed to drain for at least fifteen (15) minutes between tests. At the commencement of the next test, there shall be pressure applied for five minutes without the application of water to ensure there is no residual water remaining in the product that could affect the test results. If residual water is observed after the application of pressure, then additional drainage time shall be allowed prior to commencement of the next test.

For products segmented into multiple test specimens due to size, the evaluation of water penetration shall be confined to the respective area subject to testing.

7.3.4.1 During the test period the water source provided to the spray rack shall be regulated to the same pressure used for calibration of the spray rack. The distance from the spray rack to the test specimen shall be within ± 50 mm (± 2 in) of the spacing used for calibration. In addition, the pressure gauge shall be located at the same position in the spray rack that was used for calibration.

7.3.4.2 If the water source is not capable of achieving calibration pressure, appropriate steps shall be taken to increase the source pressure. If the water pressure cannot be increased to the pressure used during calibration it shall be noted in the test report along with the reason that the water pressure could not be achieved and that testing using a water pressure lower than the calibrated water pressure is a deviation from the ASTM E1105 procedure.

7.3.4.3 If the rack to specimen spacing or pressure gauge location cannot be duplicated, a recalibration of the rack in accordance with ASTM E 1105 is required.

7.3.5 With all operable portions of the specimen closed and locked, the specimen shall be subjected to a water penetration test in accordance with ASTM E 1105 with continuous pressure and water application. Observe and note all points of water penetration, if any, that occur during the test. If the origin of the water leakage cannot be definitively determined, a forensic evaluation shall be performed using the procedures outlined in AAMA 511 while maintaining the test pressures defined in the field testing specifications and employing the test methods defined in AAMA 503.

7.3.6 Unless defined otherwise in project specifications, water penetration shall be defined as follows:

- a) **Water penetration attributable to the surrounding wall conditions:** the presence of water not contained within drained flashing, gutters and sills which did not originate from the fenestration product or the joint between the fenestration product specimen and the wall/roof.
- b) **Water penetration attributable to the fenestration product specimen:** per ASTM E1105 with the following modification: the collection of up to 15 ml (0.5 oz) of water on an interior horizontal framing member surface in the 15- minute test period shall not be deemed a failure (see Figures 5 and 6).
- c) **Water penetration attributable to the perimeter joint:** water not controlled by a water management system that indisputably originates at the perimeter joint.

NOTE 10: If leakage through frames, frame corners, exterior perimeter seals or adjacent materials is not visible from the interior and such leakage could appear on the floor below the specimen being tested, the lower floor should be inspected for signs of leakage. A means to view, observe, or otherwise identify test-related water penetration beyond the planes of water penetration, should be accommodated if desired by any responsible party, provided that the appropriate parties are notified that repairs must be made.



FIGURE 5: Example #1 of the accumulation of 15 ml (1/2 oz) of water



Figure 6: Example #2 of the accumulation of 15 ml (1/2 oz) of water

7.3.7 If any of the specimen(s) do not conform to the prescribed air leakage or water penetration resistance requirements, or both, the manufacturer and/or the installer shall be afforded the opportunity to perform a site inspection and determine the reason for non-compliance. Non-compliant specimen(s) shall be repaired as required and retested as soon as practical (See Note 3). The remedial work shall be recorded and approved by the specifying authority, architect and/or owner. Upon satisfactorily passing a retest, the remedial work performed shall become punch list items to randomly check for similar conditions on the remainder of the project. When the inspection and remediation of a specimen requires removal and re-installation of parts or the whole specimen, it shall be at the sole discretion of the lab/testing agency to determine the feasibility of re-testing the same specimen.

NOTE 11: Retesting prior to the completion of remedial work, including full sealant cure, may render the test results inconclusive.

8.0 CALIBRATION AND VALIDATION

8.1 At least every six (6) months, the pressure measuring apparatus shall be calibrated to measure the test pressure difference within a tolerance of $\pm 2\%$ or ± 0.01 in. of water column (± 2.5 Pa), whichever is greater.

8.2 Guidelines for Validation of Air Infiltration Equipment Used for On-Site Testing

8.2.1 A full validation shall be conducted under controlled conditions in accordance with the tolerances in ASTM E283/E283M Section 9 for each air flow measurement system used for on-site testing and utilize the validation apparatus described in Figure 7 of this document. Controlled validation shall be performed at least every six (6) months.

NOTE 12: ASTM E283/E283M is referenced because validation requirements in ASTM E783 were still under development at the time this version of AAMA 503 was being developed.

8.2.2 The full validation shall consist of a minimum of three flow standards (verification plates) between 7.5 CFM and at least 20% of the equipment capacity using the validation apparatus described in Figure 7.

8.2.3 Onsite Spot Check: To ensure there is no change in measurement accuracy as a result of transport, a spot check shall be made at the beginning of each day of testing as follows:

8.2.3.1 Attach the air flow metering system and identical piping scheme to be used for testing to the validation apparatus shown in Figure 7. Determine the measured airflow through flow standards of 7.5 CFM and a point within 20% of the equipment capacity (verification plates) at the specified pressure for the flow standards.

8.2.3.2 If the results for both verification standards are within the allowable tolerance for accuracy as defined in ASTM E783, testing may proceed.

8.2.3.3 If either of the results are not within the allowable tolerance of accuracy established in ASTM E783, identify and correct the cause of the discrepancy. Verify the discrepancy has been corrected by completing a full verification in accordance with Section 8.2.1.

8.2.4 Records of controlled validation and onsite spot-checks shall be maintained for a minimum of four (4) years. Records of the most current validation and onsite spot check shall be made available for inspection at the jobsite prior to the commencement of testing. A statement that the spot check(s) was conducted was conducted prior to testing shall be part of the test report.

8.3 Guidelines for Calibration of Water Spray Equipment Used for On-Site Testing

8.3.1 A full calibration shall be conducted under controlled conditions in accordance with ASTM E1105 on the maximum height and width spray rack. Controlled calibration shall be performed at least every six (6) months.

8.3.2 The water source pressure shall be recorded during calibration using a calibrated pressure gauge that will remain a part of the spray rack throughout the six-month calibration period.

8.3.3 The spacing between the nozzles and the front plane of the catch box shall be recorded as part of the calibration data.

8.3.4 Records of controlled calibration shall be maintained for a minimum of four (4) years. Records of the most current calibrations shall be made available upon request for inspection.

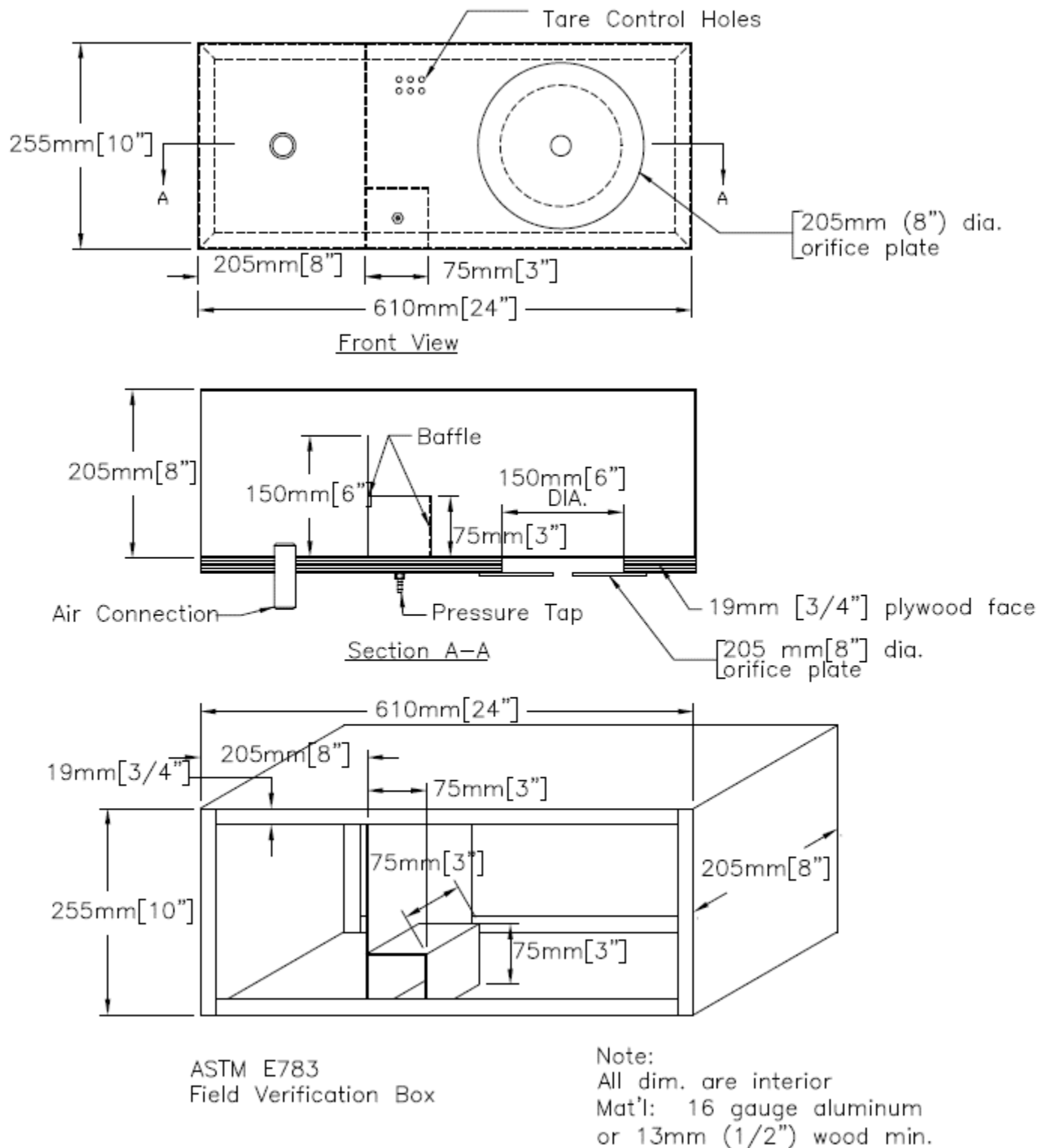


Figure 7: Validation Apparatus

9.0 TEST REPORTS

9.1 The report shall include enough information to reproduce the test. At a minimum, the following information shall be included:

9.1.1 General

Record all of the following that apply:

- The testing agency,
- Name of the individual(s) performing the tests and the FenestrationMaster® in responsible charge;
- Name, title and company affiliation of those witnessing all or part of the test(s);

NOTE 13: *This list is meant to include individuals with significant interest in the test or official observers.*

- Date and time of test, date of report, identification and location of the building shall be identified;
- The date of the last equipment calibration and the location of calibration records shall be included in the report and;
- If testing per ASTM E783 is performed, a statement that the spot check(s) required by Section 8.2.3 were conducted prior to testing.

9.1.2 Glazing Product Description

Record all of the following that apply:

- The manufacturer, model, operation type, dimensions, materials, etc.;
- Identification and location of specimen(s) within the building;
- Physical condition of all observable elements of the specimen including the weep or water management system;
- Description of any modifications made to the specimen material to the test;
- Number of retests, etc.;
- The plumb, level and square condition of the tested specimen.;
- A statement documenting the inspection of the wall construction and other conditions adjacent to the test specimen;
- Other details relevant to the installation of the tested specimen and;
- Photographs of interior and exterior views of the test specimen and any other details that could affect the test results.

9.1.3 Sampling Procedures

If applicable, describe or list the procedures established in Section 6.0.

9.1.4 Test Parameters

Record all of the following that apply to the application of the test(s):

- Reference to project specification(s) and test standards used to determine procedures, test pressure(s) and pass/fail criteria,
- List or describe the specified static pressure differential(s) used in the test,
- Chamber description, including whether the chamber was affixed to the interior or exterior of the wall/roof, and a detailed description (to include sketches or photos showing location, if appropriate) of the chamber attachment to the specimen.
- A description of any trim or wall cladding, or other materials removed during the test to provide a path for observation or to facilitate chamber attachment.
- A description of relative ambient conditions and any measures that were taken to control ambient conditions.
- Clearly identify any elements of the specimen that were not tested.
- Identify all intended areas or elements to be isolated, including techniques utilized.
- A list of any changes, modifications or adjustments made to test specimens either prior to or during the testing sequence.
 - Document any adjustments/repairs made after a failing test and prior to retesting.
- A statement confirming that:

- o The sample was inspected immediately prior to the test or installation of the chamber if it conceals portions of the fenestration assembly,
- o The original conditions were observed and documented,
- o All surfaces were dry, such that water observed during or after testing was produced by the test itself and no other possible source.

9.1.5 Test Results

Record the following:

- Actual and allowable air leakage for the product specimen.
- Actual and allowable water penetration for the product specimen
- Actual and allowable water penetration for the perimeter condition.
- Environmental conditions as measured at the time of the test, including:
 - o Wind speed, wind direction,
 - o Precipitation,
 - o Barometric pressure
 - o Ambient temperature.
- Photos that support the description of the test specimen and the test results

9.1.6 Additional Observations

If problems with a specimen installation are observed, they shall be brought to the responsible contractor's and/or manufacturer representative's attention.

The observations to be recorded shall include but not be limited to the following:

- Deterioration of building elements due to water penetration
- Deviations of the installation from the drawings of record
- Staining or discoloration of building components
- Evidence of damage to either the installed product or the surrounding building elements
- Unusual or unexpected evidence of water penetration or air leakage which would require remediation
- Any observed performance or installation details which might be deemed of importance to a subsequent forensic investigation.

9.1.7 Compliance Statement

Make a statement that the tests were conducted in accordance with this specification or completely describe any deviation, including the purpose and the technical rationale for the deviation.

Changes from AAMA 503-08 to AAMA 503-08(2014)

- Various editorial changes were made.
- Specifier Notes to Short Form Field Testing Specification were expanded to provide additional guidance.
- Added Section 5.1.3 to provide direction on availability of applicable project specifications and standards.
- Updated Section 6.0 Sampling, Site Preparation and Responsible Parties.
- Updated test procedures so that air leakage resistance testing is no longer optional (Section 7.1).
- Added new Section 8.0 Calibration and Validation.
- More figures were added to the document for user guidance.
- This document was updated by the Testing Criteria of Commercial Fenestration Updates Task Group.

Changes from AAMA 503-08 to AAMA 503-08(2014)

- Various editorial changes were made



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